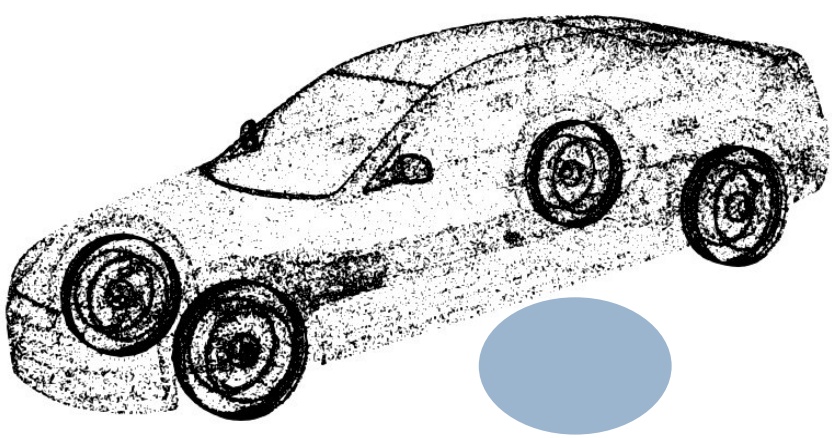
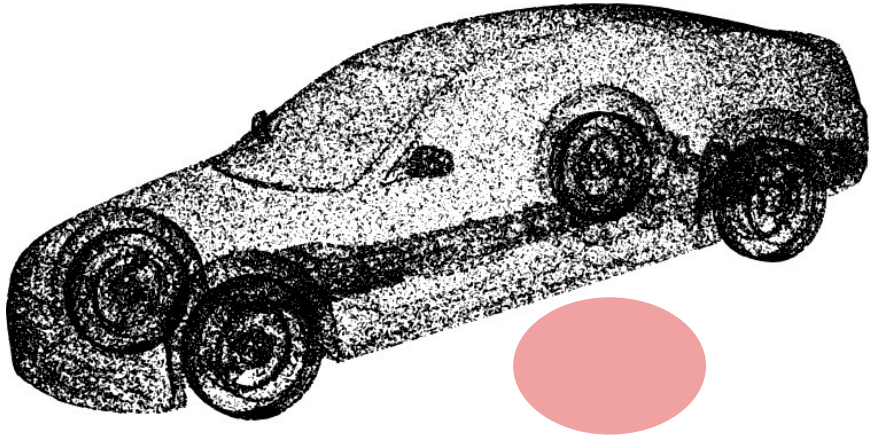
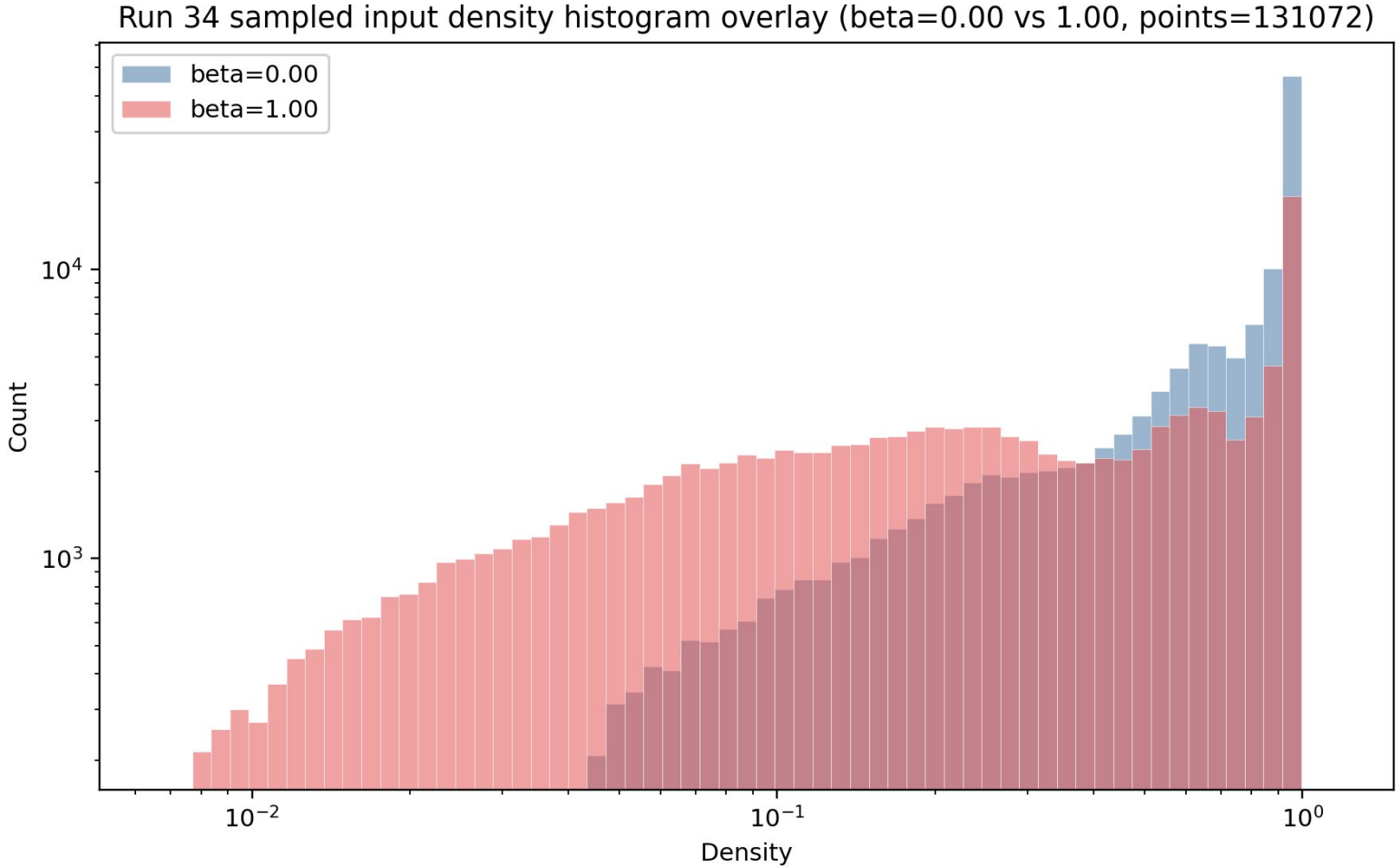
Meeting 15.06.2026

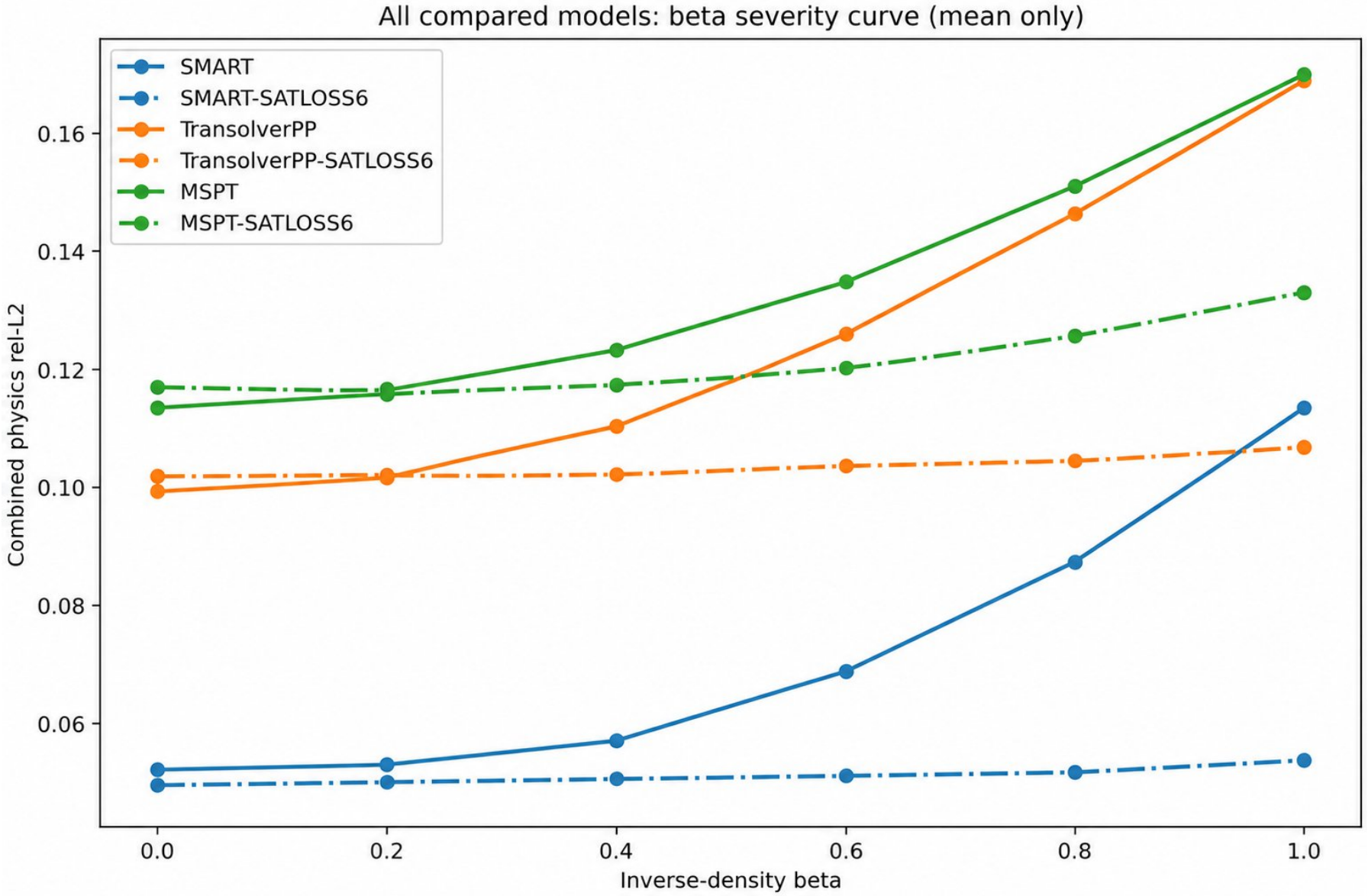
Update with Nils and Parsa



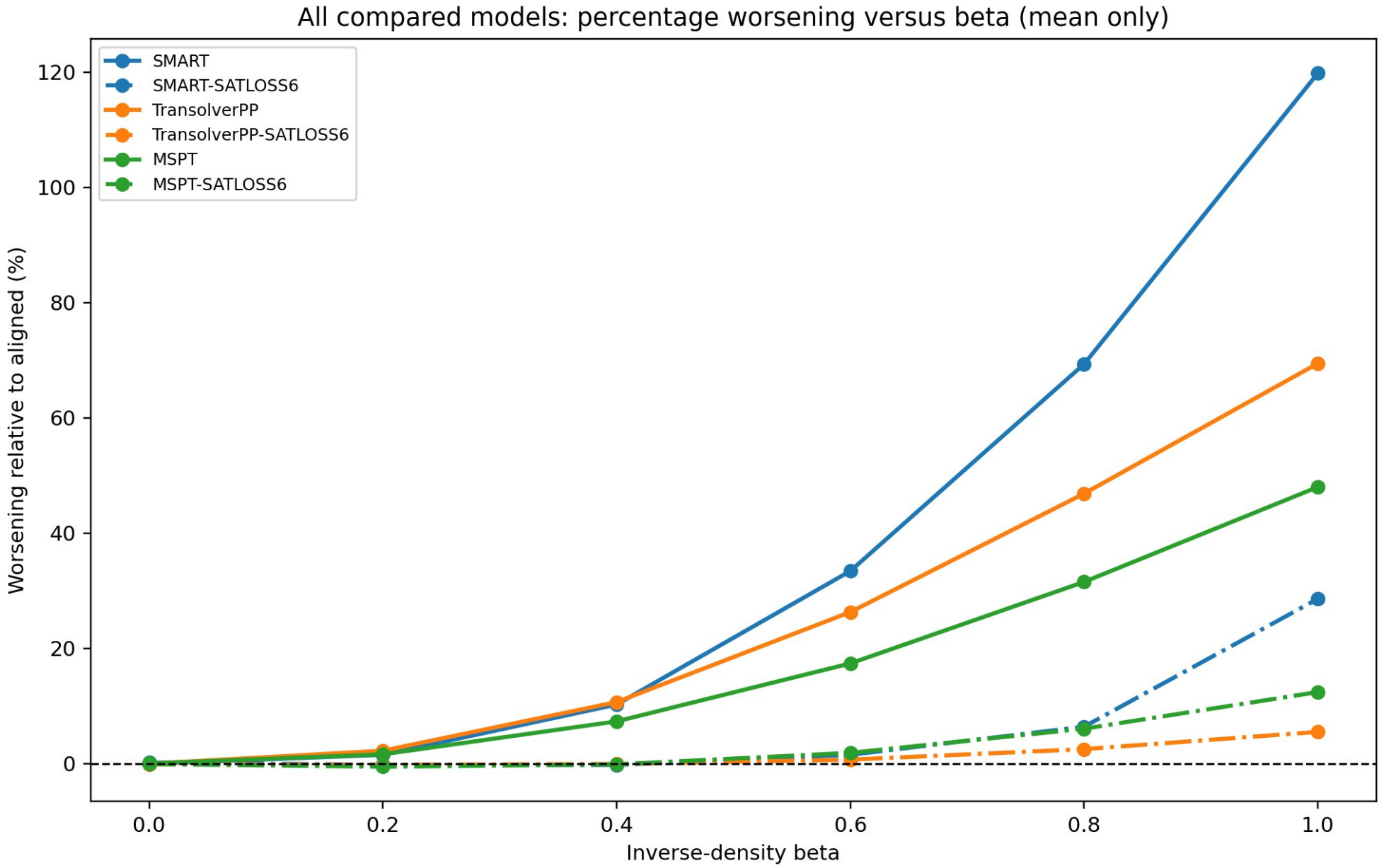



Run 34 sampled input density histogram overlay (beta=0.00 vs 1.00, points=131072)
beta=0.00
beta=1.00
Count
10^4
10^3
10^{-2}
10^{-1}
10^0
Density

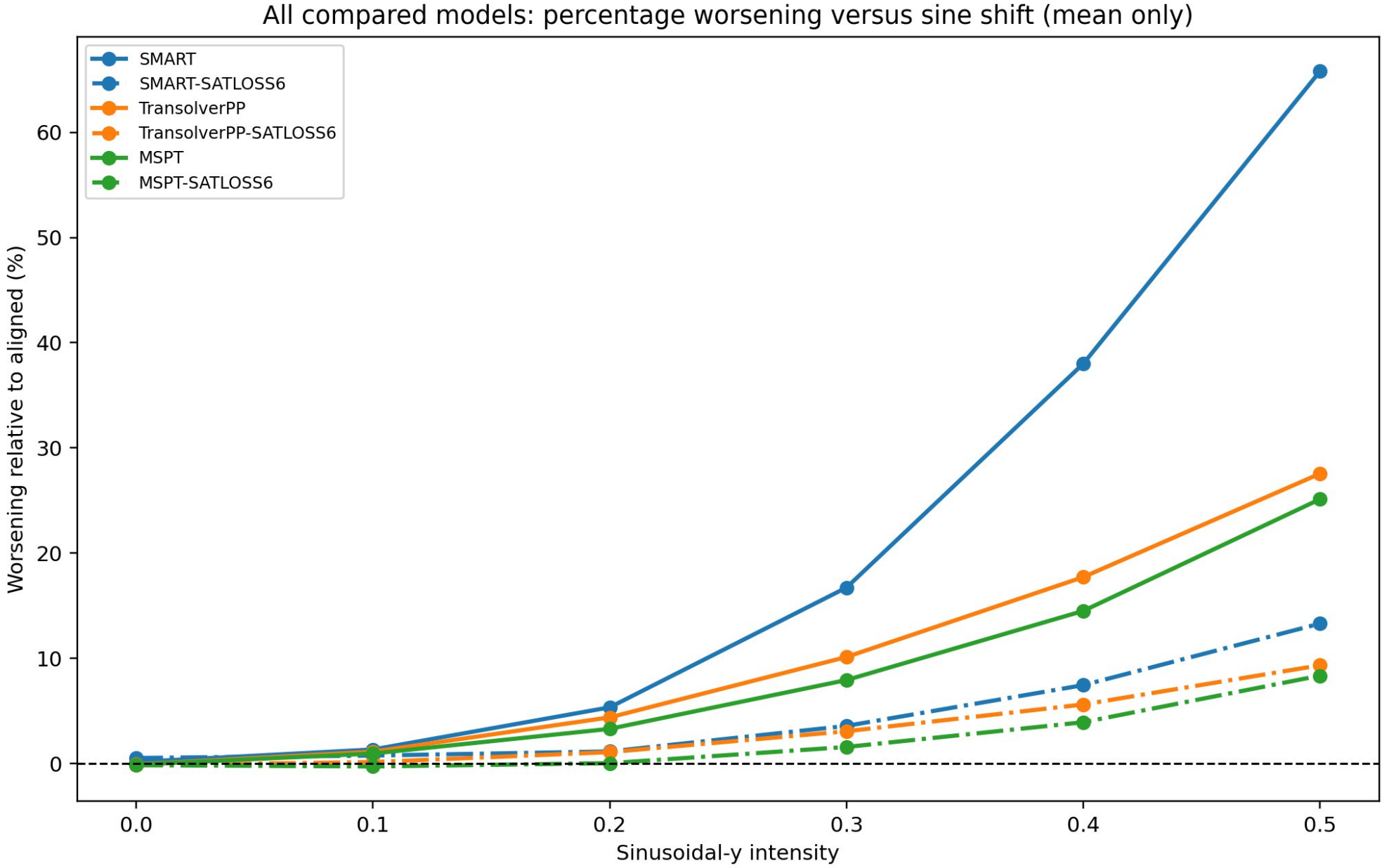
Trying different models – beta test



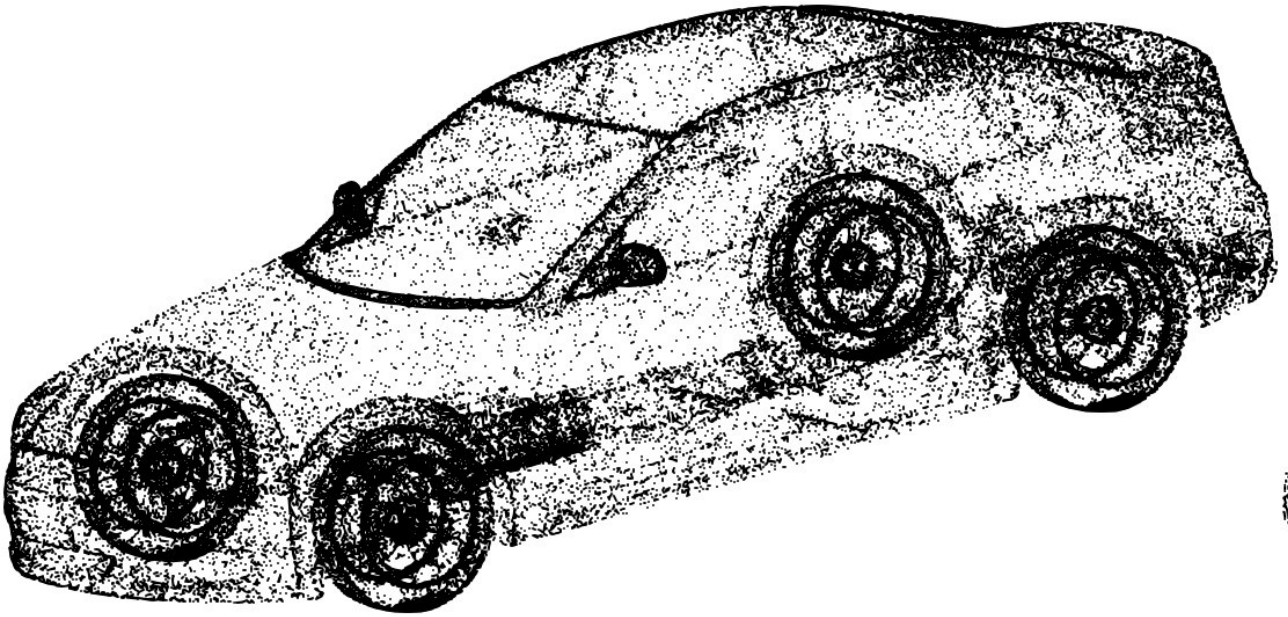
Trying different models – beta test (relative to beta 0)

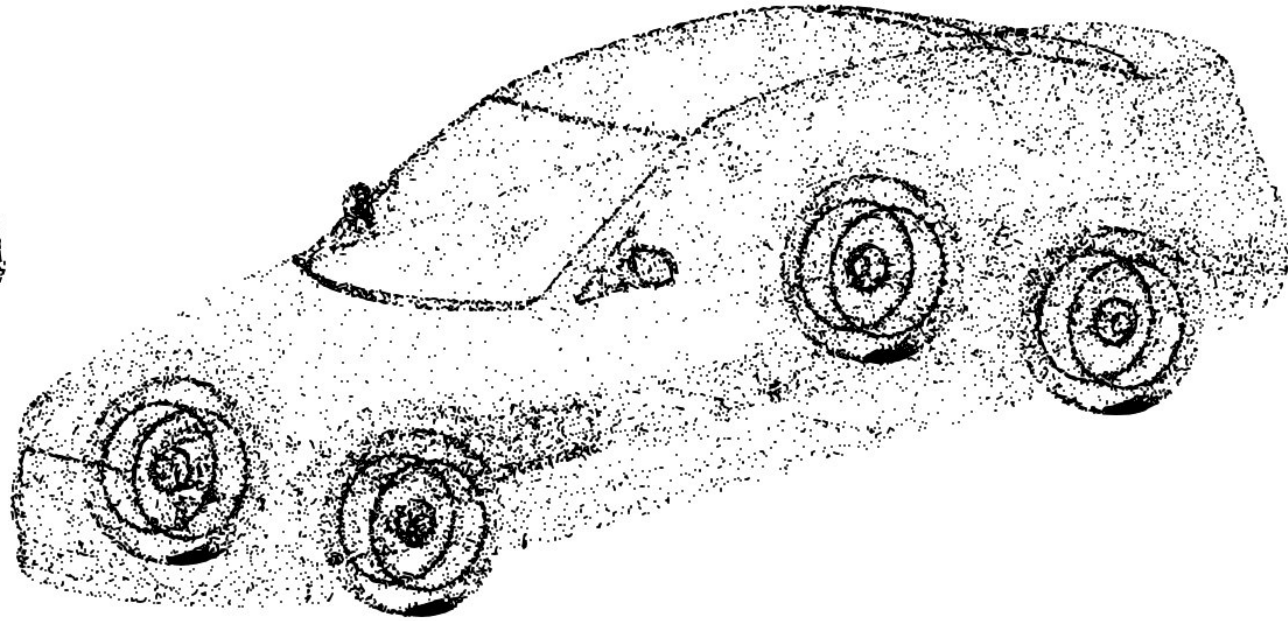


Trying different models – sine test (relative to sine 0)



Previous methodology in the literature: Random mask



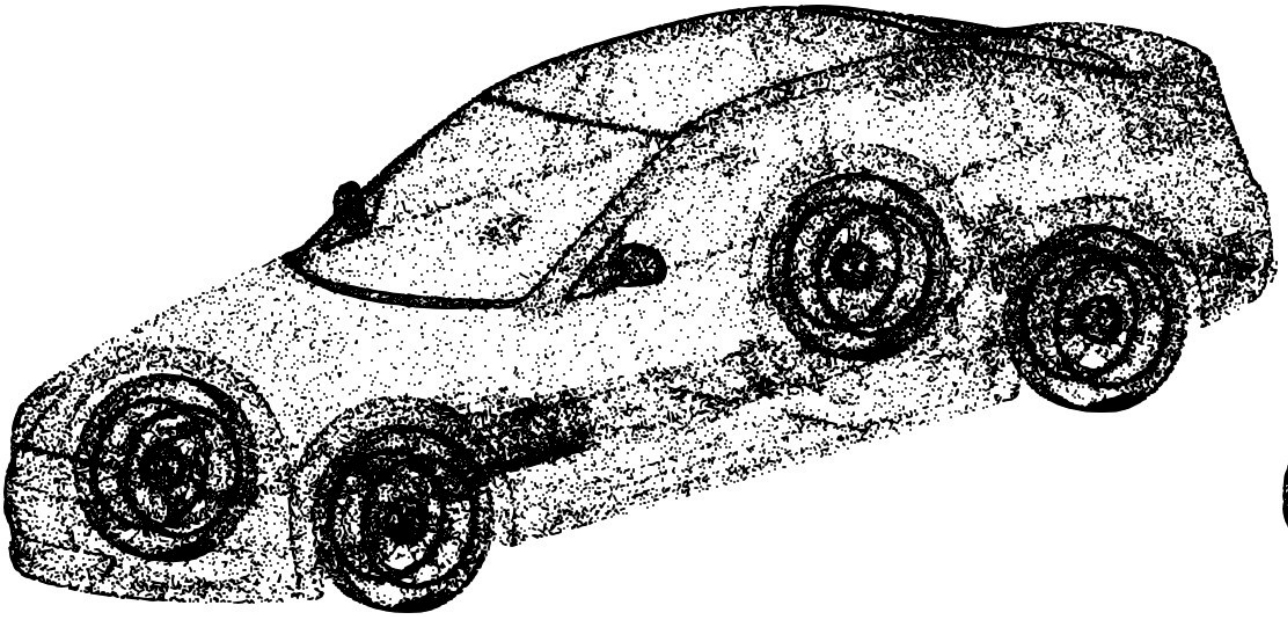


131k points - view 1

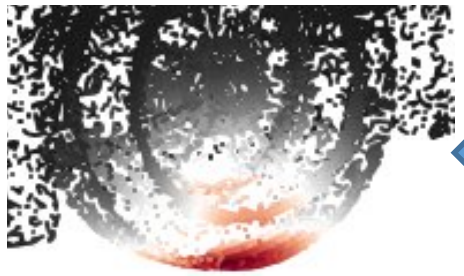
33k points - view 2

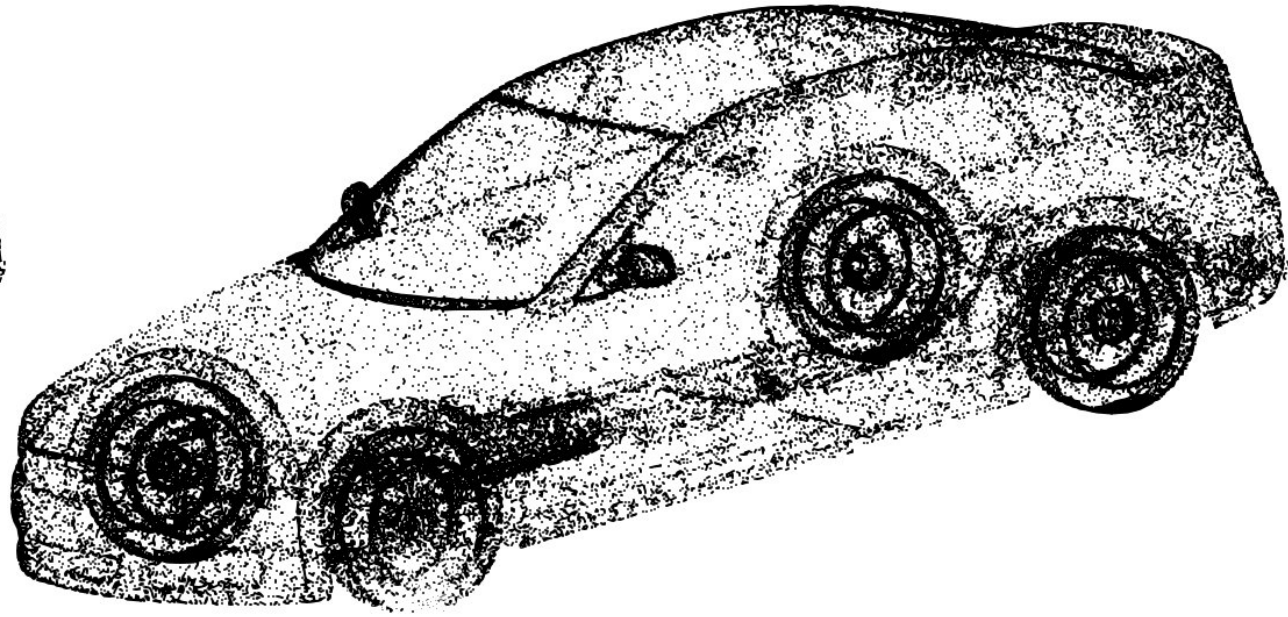
Prediction match loss

Previous methodology in the literature: Gaussian Ball Mask



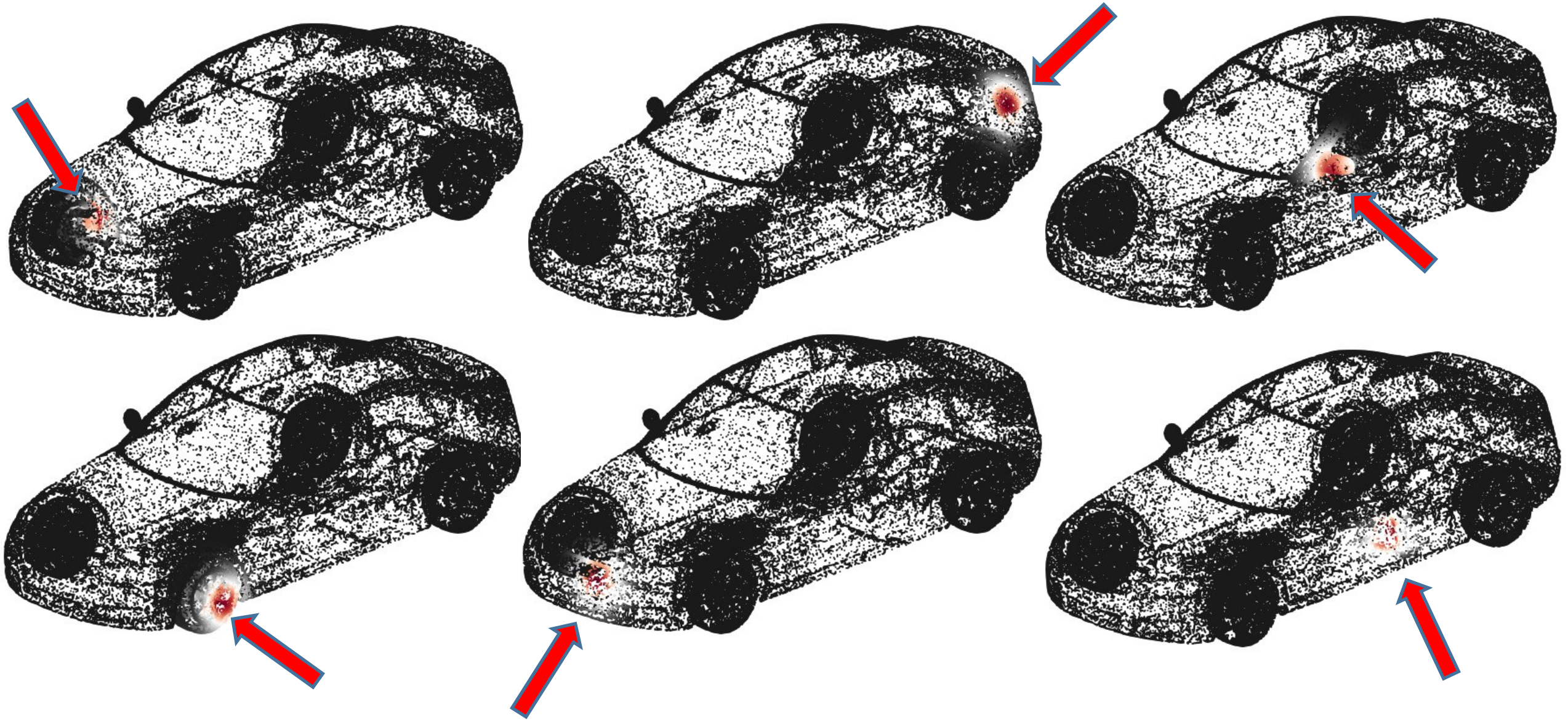
131k points - view 1



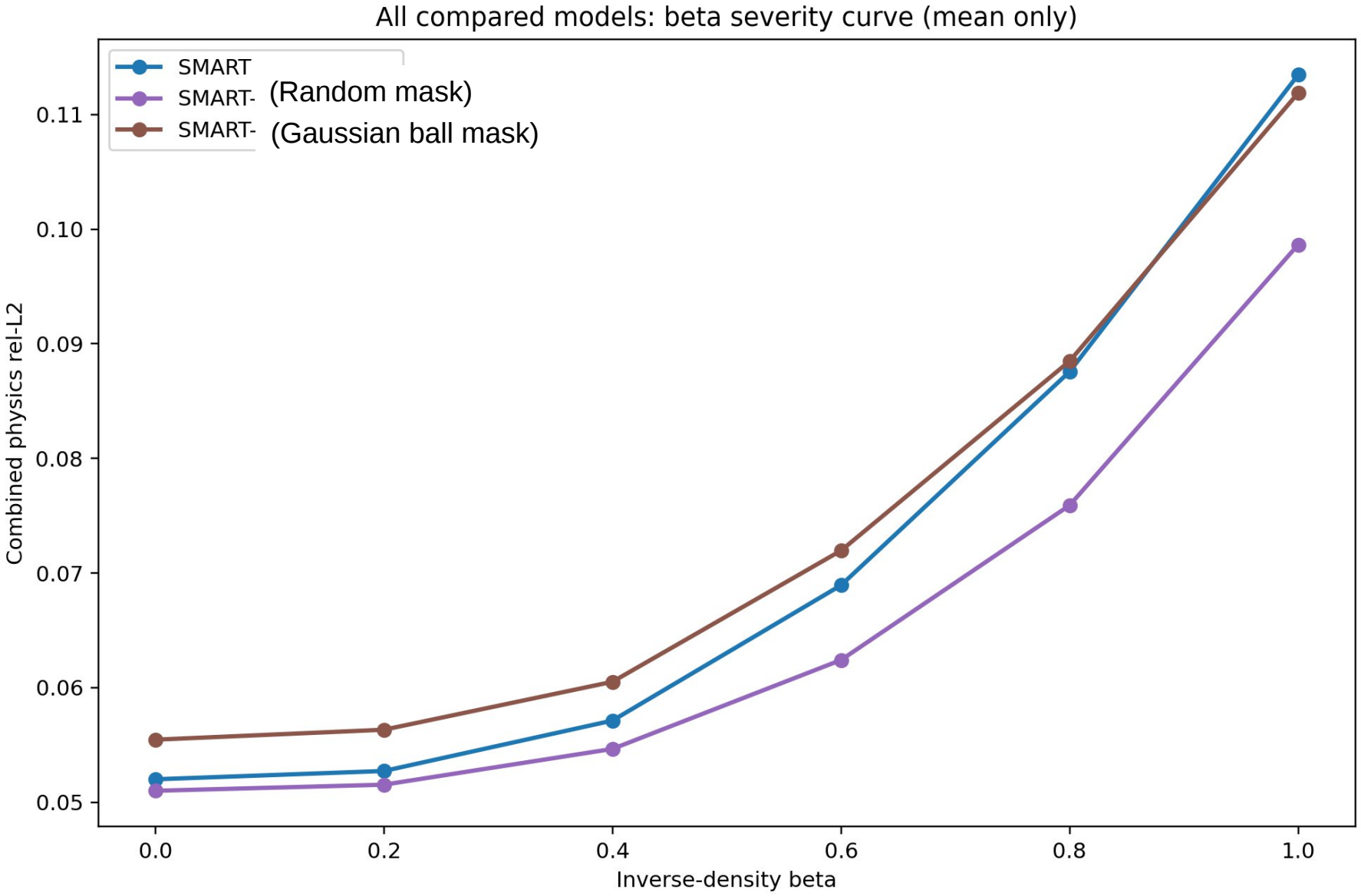


127k points - view 2

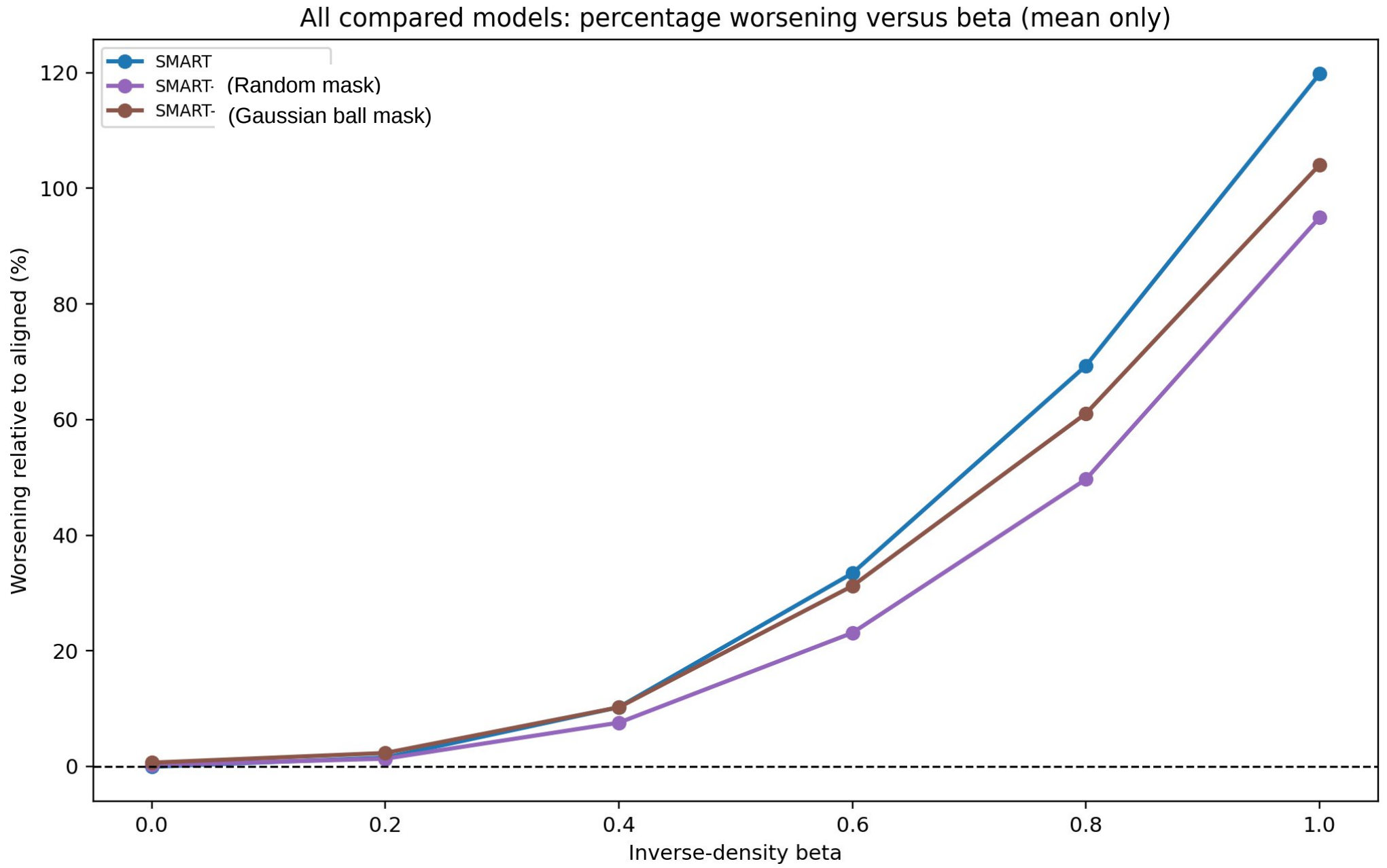
Point sampling Baselines – Gaussian Ball Mask



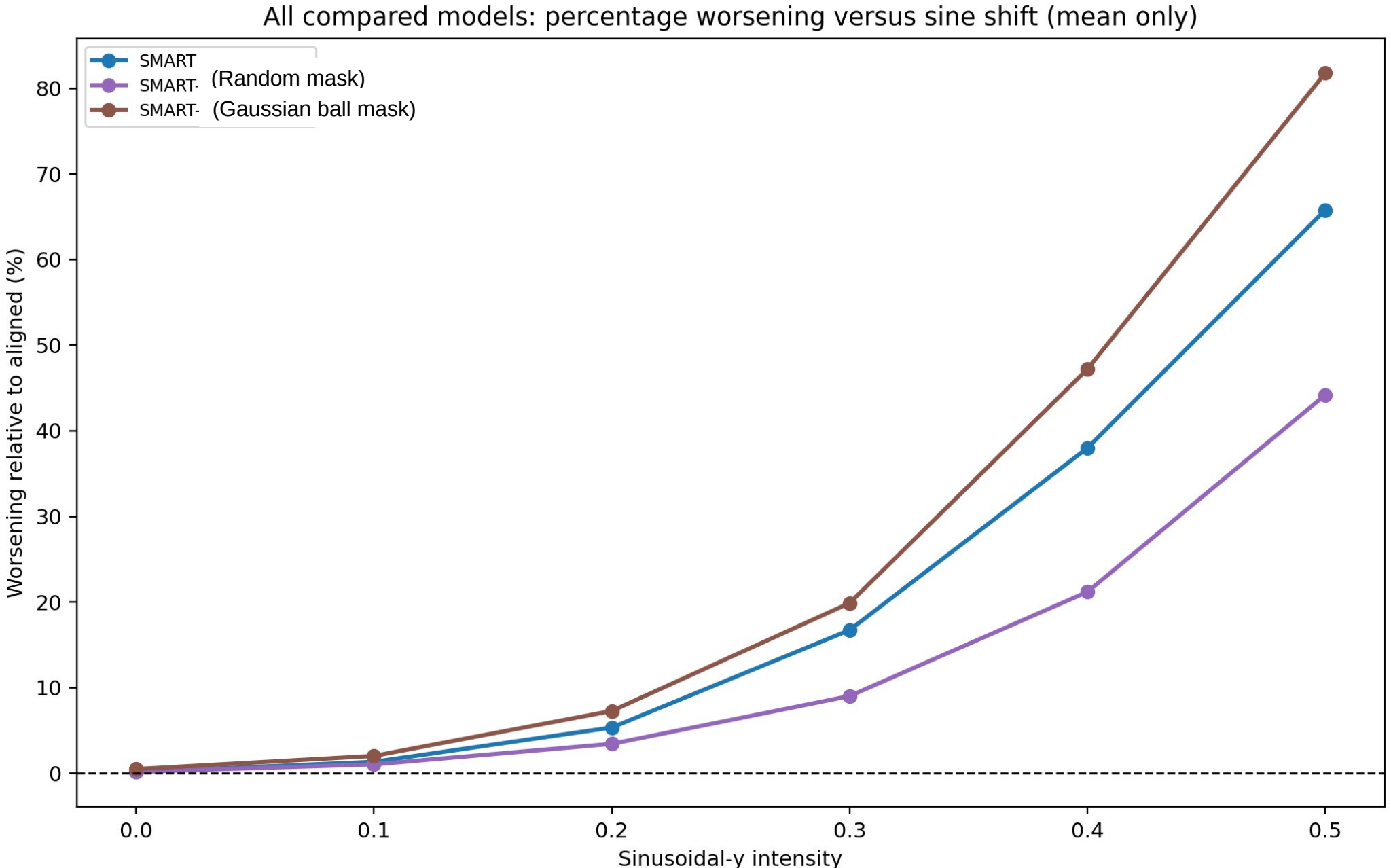
Result of baseline methods - beta



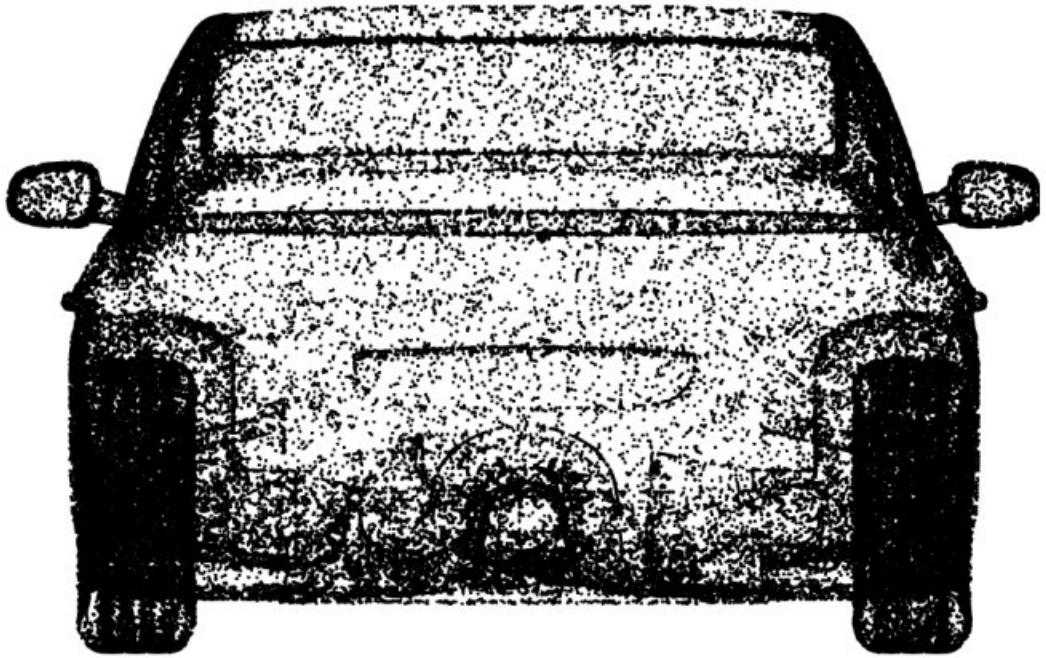
Result of baseline methods- beta (relative to beta 0)

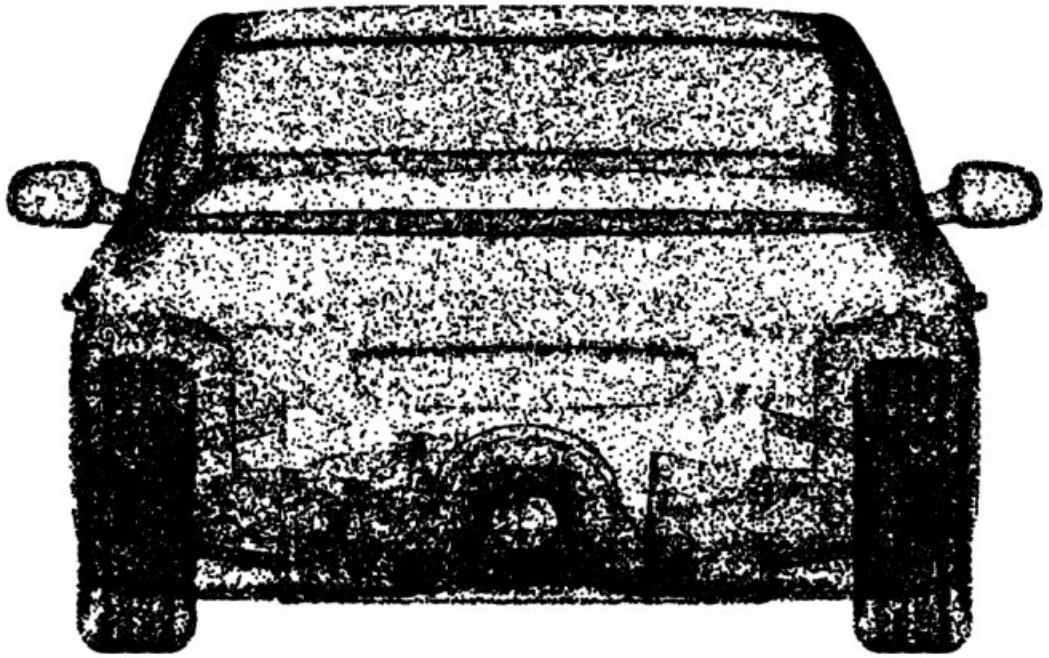


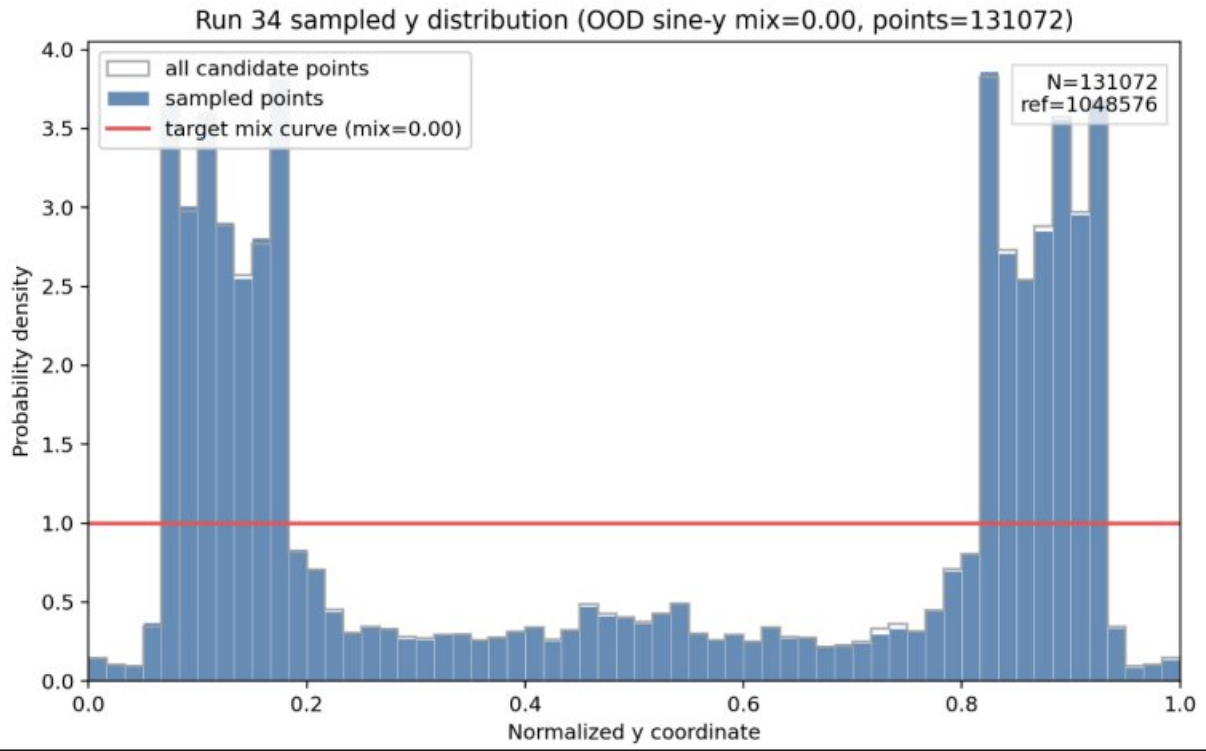
Result of baseline methods- beta (relative to beta 0)

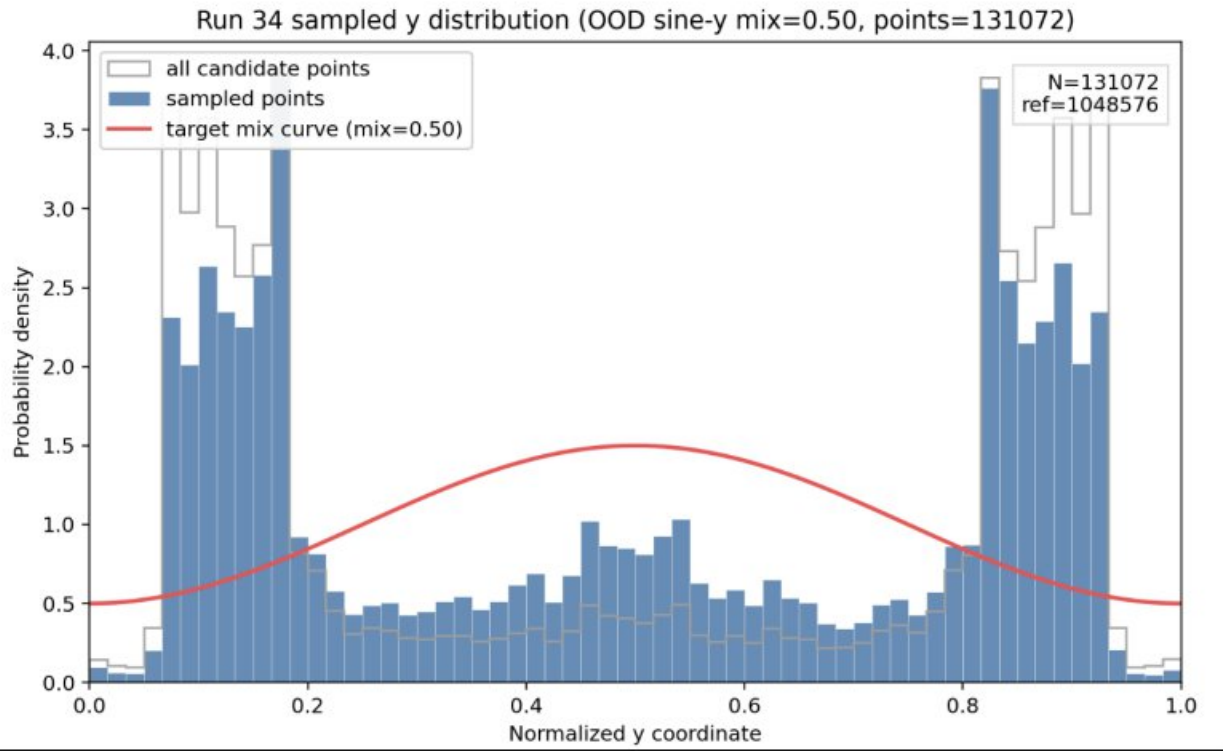


Backup



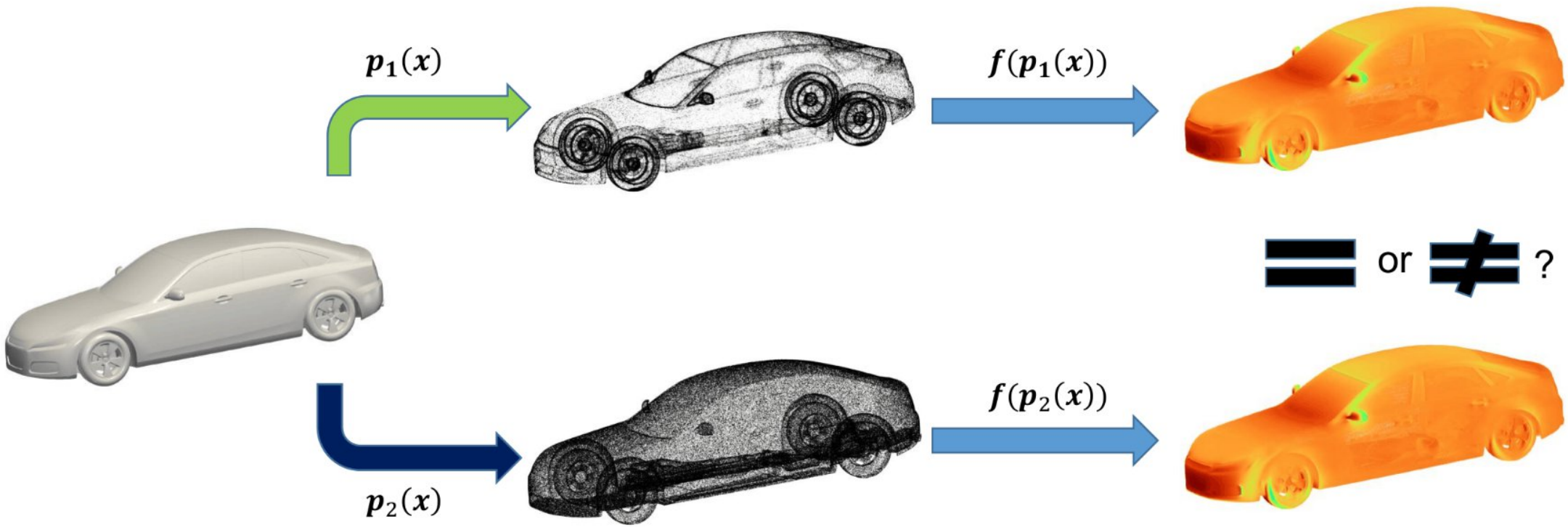



Run 34 sampled y distribution (OOD sine-y mix=0.00, points=131072)
all candidate points
sampled points
target mix curve (mix=0.00)
N=131072
ref=1048576
Probability density
Normalized y coordinate
0.0
0.5
1.0
1.5
2.0
2.5
3.0
3.5
4.0
0.0
0.2
0.4
0.6
0.8
1.0

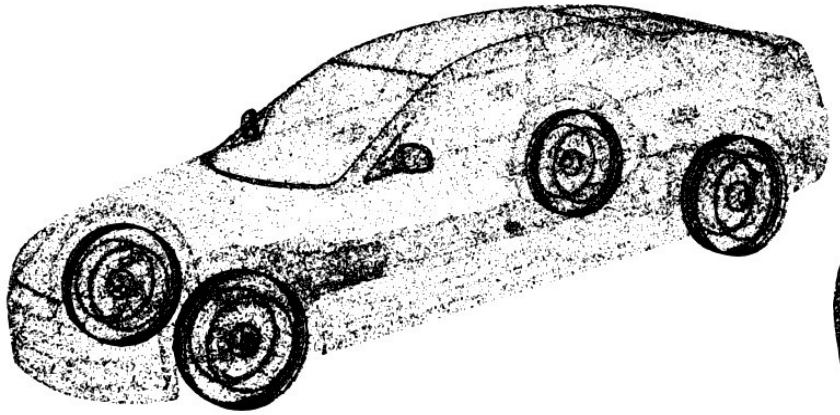

Run 34 sampled y distribution (OOD sine-y mix=0.50, points=131072)
all candidate points
sampled points
target mix curve (mix=0.50)
N=131072
ref=1048576
Probability density
Normalized y coordinate
0.0
0.5
1.0
1.5
2.0
2.5
3.0
3.5
4.0
0.0
0.2
0.4
0.6
0.8
1.0

Motivation

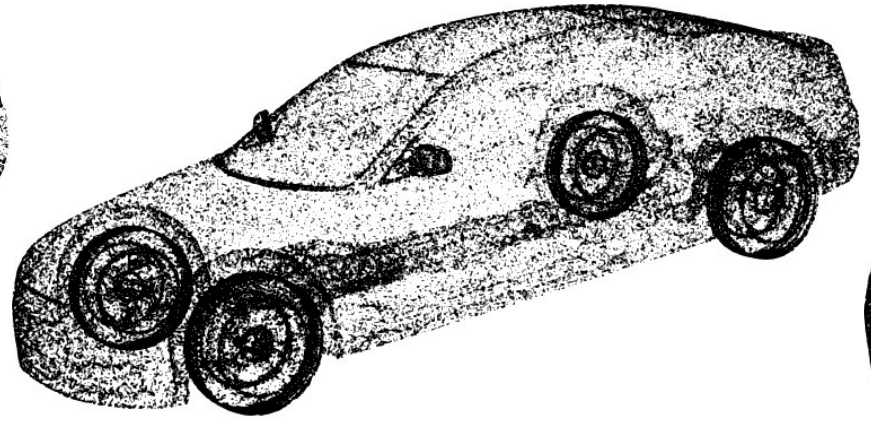
- When we are making prediction y for an input geometry x , our prediction is expected to be $\boldsymbol{f}(\boldsymbol{x}) = \boldsymbol{y}$. However, geometries are first preprocessed into a pointcloud using $p(x)$ and then they are fed to the model. So what we get is actually $\boldsymbol{f}(\boldsymbol{p}(\boldsymbol{x}))$. Not having access to the $\boldsymbol{p}(\boldsymbol{x})$ poses a challenge for an ideal prediction if the network is not agnostic to how behaves.
- We need to test whether for two different sampling strategy $\boldsymbol{p_1}(\boldsymbol{x})$and $\boldsymbol{p_2}(\boldsymbol{x})$or not.



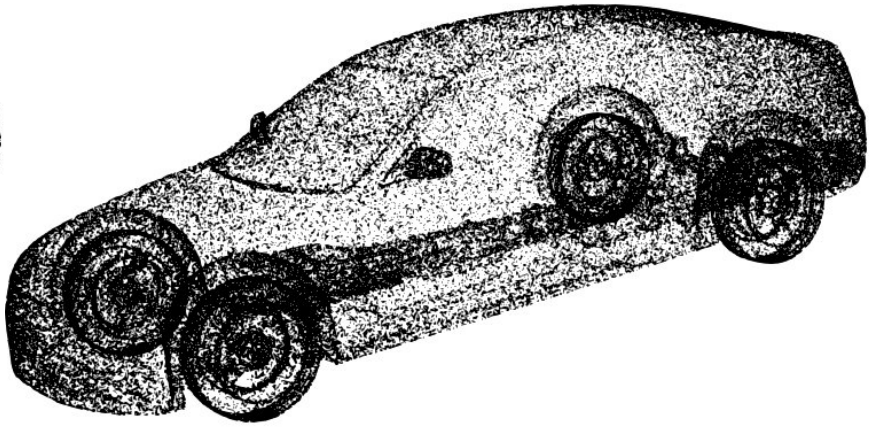
The density-based point sampling



random distribution β=0



Density correction intensity β=0.5



Density correction intensity β=1.0

The algorithm first estimates a **log-density** for each point based on its local k-nearest neighbors (typically k=24):

$$\log \rho_j = -\log(d_k^{(j)})$$

Once the log-densities are calculated for all points, the sampling weights and final probabilities are derived:

$$w_i = \rho_i^{-\beta}$$

$$p_i = \frac{w_i}{\sum_j w_j}$$